

JAVA SDE-2 INTERVIEW PREPARATION SERIES

SERIES INDEX - 4 SEGMENTS / 41 BOOKS

# Java SDE-2 Interview Preparation Series Index

A Basics-to-Advanced Navigation Guide

BY

**Vinay Reddy Kalluri**

Choose Java, DSA, Frameworks, or System Design, then follow a clear prerequisite-aware path within the segment.

JAVA | DSA | FRAMEWORKS AND DATA | SYSTEM DESIGN

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

Series edition - Updated 2026-08-02

Open educational interview-preparation edition

# Choose Your Learning Segment

This library has four clear paths: Java Engineering, Data Structures and Algorithms, Frameworks/Data/Messaging, and System Design. Choose the path that matches your immediate goal, then follow its numbered books in order. Frameworks and System Design intentionally build on the Java foundation.

Keep the canonical [00 - start - here](#) through [04 - system - design](#) folders together so relative links can work in viewers that permit local-file navigation. The printed series codes remain the fallback when a viewer blocks those links.

## Choose your starting point in 60 seconds

**Start rule:** select one segment first. Inside it, begin with Book 01 unless you can pass that book's readiness checks without notes.

| Your goal                             | Start here                                     | Continue with                                                                                    |
|---------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Learn or rebuild Java                 | JAVA 01 - Java Foundations                     | Continue through the Java segment in order                                                       |
| Build interview problem-solving skill | DSA 01 - Time and Space Complexity             | Continue through the DSA segment in order                                                        |
| Prepare for Java backend interviews   | FW 01 - MySQL                                  | Continue through persistence, Spring, data stores, and messaging                                 |
| Prepare for architecture rounds       | SD 01 - Design, Backend, Testing, and Security | Continue with distributed systems, then the generative-AI design round, after the framework path |

## The learning loop for every volume

- 1 **Recognize:** use the entry signals and decision maps to identify the pattern.
- 2 **Understand:** restate the contract, invariant, and failure modes in your own words.
- 3 **Implement:** type the Java example and test boundaries before looking back.
- 4 **Explain:** give the complexity, trade-offs, and one production follow-up aloud.
- 5 **Prove readiness:** pass the completion check before opening the next volume.

## Part 1 - Java Engineering

Start with the language, developer workflow, build tools, core libraries, JVM, concurrency, performance, and Java interview revision.

### JAVA 01 - Java Foundations for Problem Solving

This volume is the canonical Java language foundation and the recommended first book in the series. It starts with source-to-execution and program structure, builds through types, control flow, methods, arrays, strings, objects, core libraries, exceptions, generics, and usage-first collections, then converts that knowledge into interview-quality implementation and debugging skill.

Open [Java-SDE2-JAVA-01-Java-Foundations-for-Problem-Solving.pdf](#).

### JAVA 02 - Git and GitHub for Java Engineers

Build dependable Git and GitHub judgment from local snapshots through pull requests, protected branches, secure Java automation, releases, and production recovery.

Open [Java-SDE2-JAVA-02-Git-and-GitHub.pdf](#).

### JAVA 03 - Maven and Gradle for Java Engineers

Teach one durable Java build model first, map Maven and Gradle to it precisely, and develop the diagnostic and delivery judgment expected in SDE-2 interviews and production ownership.

Open [Java-SDE2-JAVA-03-Maven-and-Gradle.pdf](#).

### JAVA 04 - Language, OOP, and Modern Java

This book covers the language-engineering depth expected when SDE-2 interviews move beyond coding patterns into API design and type safety.

Open [Java-SDE2-JAVA-04-Language-OOP-and-Modern-Java.pdf](#).

### JAVA 05 - Collections, Streams, and I/O

This Java-library book complements the DSA shelf by emphasizing API contracts, representation costs, resource ownership, and production boundaries.

Open [Java-SDE2-JAVA-05-Collections-Streams-and-IO.pdf](#).

## JAVA 06 - JVM and Java Execution

This book builds the execution model needed to reason about Java behavior beneath syntax without presenting implementation details as language guarantees.

Open [Java-SDE2-JAVA-06-JVM-and-Execution.pdf](#).

## JAVA 07 - Java Concurrency and the Memory Model

This book develops concurrency correctness from first principles and then connects it to production scheduling, cancellation, backpressure, and testing.

Open [Java-SDE2-JAVA-07-Concurrency-and-Memory-Model.pdf](#).

## JAVA 08 - Performance, Diagnostics, and GC Incidents

This book teaches an evidence-first diagnostic method so performance claims and tuning actions remain falsifiable and operationally safe.

Open [Java-SDE2-JAVA-08-Performance-Diagnostics-and-GC-Incidents.pdf](#).

## JAVA 09 - Java Question Bank, Study Plan, and Reference

This book is the core Java revision system. It consolidates interview questions, selected solutions, study loops, terminology, and primary references without replacing deliberate practice.

Open [Java-SDE2-JAVA-09-Question-Bank-Study-Plan-and-Reference.pdf](#).

## Part 2 - Data Structures and Algorithms

Build complexity intuition, mathematical foundations, implementation patterns, data structures, algorithms, and problem-solving confidence.

### DSA 01 - Time and Space Complexity for Java Interviews

This volume is the bridge from Java syntax to algorithmic problem solving. It begins with concrete operation counts, introduces growth and space gently, then develops the qualified cost models, constraints, invariants, amortization, and optimization explanations expected in SDE-2 interviews.

Open [Java-SDE2-DSA-01-Time-and-Space-Complexity.pdf](#).

### DSA 02 - Number Systems and Math Foundations for DSA Interviews

This volume closes the mathematical prerequisite gap before bit manipulation, arrays, binary search, hashing, and number-based coding problems. It is practical DSA mathematics, not an academic number-theory textbook.

Open [Java-SDE2-DSA-02-Number-Systems-and-Math-Foundations.pdf](#).

### DSA 03 - Number Systems Interview Patterns and Rapid Revision

This linked workbook completes the DSA-02 numeric foundation with a 52-implementation catalog, Java number traps, conceptual and code-output questions, debugging tasks, coding exercises, solutions, follow-up chains, cheat sheet, and readiness assessment.

Open [Java-SDE2-DSA-03-Number-Systems-Interview-Workbook.pdf](#).

### DSA 04 - Bit Manipulation in Java

This dedicated volume turns the binary foundation from DSA-02 into a progressive Java interview toolkit. It begins with visible bit patterns and safe operators, derives each shortcut instead of asking readers to memorize it, then develops XOR families, subset state, SDE-2 problem patterns, Java APIs, and production-quality bit fields.

Open [Java-SDE2-DSA-04-Bit-Manipulation-in-Java.pdf](#).

### DSA 05 - Loop Mastery, Patterns, and Index Calculations

This volume is the bridge from basic Java control flow to reliable array and string problem solving. It begins with the execution model of for, while, do-while, and enhanced-for loops; builds a visual language for ranges and invariants; then derives pointer, compaction, window, search, and matrix patterns without asking the reader to memorize templates blindly.

Open [Java-SDE2-DSA-05-Loop-Mastery-and-Index-Calculations.pdf](#).

## DSA 06 - Arrays and Array Problem-Solving Patterns

Arrays are the default interview state space. This volume begins with storage, indexes, references, copying, and traversal before building a constraint-driven pattern map for reliable SDE-2 solutions.

Open [Java-SDE2-DSA-06-Arrays-and-Array-Patterns.pdf](#).

## DSA 07 - Strings and String Problem-Solving Patterns

This prerequisite-first volume takes a reader from SDE-1 String mechanics through the substring, frequency, window, parsing, and exact-search patterns expected in SDE-2 interviews. Every advanced technique follows a baseline, invariant, dry run, Java implementation, failure boundary, and complete complexity discussion.

Open [Java-SDE2-DSA-07-Strings-and-String-Patterns.pdf](#).

## DSA 08 - Hashing: Maps, Sets, Frequency, and Prefix State

Hashing is both a DSA pattern and a Java object-contract problem. This volume connects the two so expected  $O(1)$  claims remain defensible.

Open [Java-SDE2-DSA-08-Hashing-Maps-Sets-and-Prefix-State.pdf](#).

## DSA 09 - Recursion and Backtracking in Java

This volume separates recursion as a method contract from backtracking as controlled state-space search, with Java stack and mutation behavior made explicit.

Open [Java-SDE2-DSA-09-Recursion-and-Backtracking.pdf](#).

## DSA 10 - Linked Lists: Pointer Reasoning and Mutation

Linked-list problems expose whether a candidate can preserve reachability while mutating references. This volume makes those invariants visible.

Open [Java-SDE2-DSA-10-Linked-Lists.pdf](#).

## DSA 11 - Stacks, Queues, Deques, and Monotonic Patterns

This volume unifies ordering data structures around the state each one preserves, then applies monotonic structures to linear-time interview patterns.

Open [Java-SDE2-DSA-11-Stacks-Queues-Deques-and-Monotonic-Patterns.pdf](#).

## WHY IT WORKS | DSA 12 - Binary Search: Bounds, Answers, and Invariants

Binary search is a proof pattern over monotone state. This volume centers interval semantics, duplicates, insertion points, overflow-safe midpoint calculation, and search on answers.

Open [Java-SDE2-DSA-12-Binary-Search.pdf](#).

## DSA 13 - Trees, Binary Search Trees, and Tries

Tree interviews test recursion, state ownership, structural invariants, and the ability to translate ordering into pruning.

Open [Java-SDE2-DSA-13-Trees-BSTs-and-Tries.pdf](#).

## DSA 14 - Heaps, Priority Queues, Selection, and Top-K

This volume distinguishes heap order from global sorting and develops priority-based interview patterns without hiding comparator or mutation risks.

Open [Java-SDE2-DSA-14-Heaps-Priority-Queues-and-Top-K.pdf](#).

## DSA 15 - Graphs: Traversal, Ordering, Paths, and Union-Find

Graph problems become manageable once the model and edge semantics are explicit. This volume emphasizes algorithm selection, state, overflow, and correctness.

Open [Java-SDE2-DSA-15-Graphs.pdf](#).

## WHY IT WORKS | DSA 16 - Greedy Algorithms: Recognition, Proofs, and Scheduling

Greedy solutions are short to code and hard to justify. This volume centers the proof obligation, not the superficial loop.

Open [Java-SDE2-DSA-16-Greedy-Algorithms.pdf](#).

## DSA 17 - Dynamic Programming: State, Transitions, and Optimization

This volume makes dynamic programming a specification process: define exactly what a state means, how it depends on smaller states, and what information can be compressed.

Open [Java-SDE2-DSA-17-Dynamic-Programming.pdf](#).

## Part 3 - Frameworks, Data, and Messaging

Learn SQL, persistence, Spring, data stores, caching, and messaging in a prerequisite-correct Java backend path.

### FW 01 - MySQL for Java Backend Interviews

Establish relational and SQL fundamentals before ORM abstractions, then build the InnoDB, indexing, concurrency, JDBC, migration, and diagnostic judgment expected from a Java SDE-2 engineer.

Open [Java-SDE2-FW-01-MySQL.pdf](#).

### FW 02 - Hibernate and JPA for Java Backend Interviews

Teach ORM behavior only after relational and transaction fundamentals, then make persistence-context identity, flush, fetching, query shape, batching, locking, caching, equality, and diagnosis observable and testable.

Open [Java-SDE2-FW-02-Hibernate-and-JPA.pdf](#).

### FW 03 - Spring Framework for Java Engineers

Build a precise Java object-graph and proxy mental model before Spring Boot automation, then develop the transaction, web, testing, concurrency, and incident judgment expected from SDE-2 backend engineers.

Open [Java-SDE2-FW-03-Spring-Framework.pdf](#).

### FW 04 - Spring Boot for Java Backend Engineers

Build on Spring Framework mechanics to teach how Boot assembles, configures, tests, packages, observes, secures, and operates production Java services from beginner foundations through SDE-2 incident reasoning.

Open [Java-SDE2-FW-04-Spring-Boot.pdf](#).

### FW 05 - Spring Boot and REST APIs

This book develops the Spring and HTTP reasoning expected in Java SDE-2 interviews while keeping framework behavior tied to explicit contracts and failure modes.

Open [Java-SDE2-FW-05-Spring-Boot-and-REST.pdf](#).

## FW 06 - Spring Data for Java Backend Engineers

Build strong repository-first reasoning, query-shape confidence, transaction correctness, and performance awareness before and during SDE-2 interviews.

Open [Java-SDE2-FW-06-Spring-Data.pdf](#).

## FW 07 - MongoDB for Java Backend Interviews

Teach when document storage fits and how schema, indexing, consistency, and access patterns drive the design.

Open [Java-SDE2-FW-07-MongoDB.pdf](#).

## FW 08 - Redis for Java Backend Interviews

Teach Redis as a distinct system with explicit consistency and failure boundaries, not merely a fast map.

Open [Java-SDE2-FW-08-Redis.pdf](#).

## FW 09 - Persistence, SQL, JPA, and Caching

This book develops the persistence judgment needed to design correct, measurable Java data paths instead of treating databases, ORMs, and caches as interchangeable storage APIs.

Open [Java-SDE2-FW-09-Persistence-SQL-and-Caching.pdf](#).

## FW 10 - Apache Kafka and Spring Kafka

Keep Apache Kafka semantics and Spring Kafka integration together while teaching the platform model before framework abstractions.

Open [Java-SDE2-FW-10-Apache-Kafka-and-Spring-Kafka.pdf](#).

## FW 11 - Spring Ecosystem Extensions

Teach the major Spring extensions from prerequisites and runtime mechanics through SDE-2 trade-offs without implying every service needs every module.

Open [Java-SDE2-FW-11-Spring-Ecosystem-Extensions.pdf](#).

## FW 12 - Spring AI for Java Engineers

Place Spring AI after the core backend and Spring foundations so AI integration is taught as production software engineering.

Open [Java-SDE2-FW-12-Spring-AI.pdf](#).

## Part 4 - System Design

Move from component boundaries to distributed-system reasoning after the Java backend foundations are secure.

### SD 01 - Design, Backend, Testing, and Security

This book covers the engineering judgment that distinguishes SDE-2 candidates from syntax-only candidates: boundaries, invariants, failure policy, observability, and maintainability.

Open [Java-SDE2-SD-01-Design-Backend-Testing-and-Security.pdf](#).

### SD 02 - Distributed Systems and System Design

This book applies a constraint-first system-design method spanning distributed data, messaging, failure control, and operations.

Open [Java-SDE2-SD-02-Distributed-Systems-and-System-Design.pdf](#).

### SD 03 - Generative AI System Design

This book covers the generative-AI design round that entered standard interview loops after 2024. It treats inference latency, token cost, and non-determinism as capacity constraints, and applies the same constraint-first method used for traditional distributed-system design.

Open [Java-SDE2-SD-03-Generative-AI-System-Design.pdf](#).

## Part 5 - Recommended Study Rhythm

# Build, Practice, Explain, Revisit

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For each volume, complete one learning pass, one no-notes implementation pass, one spoken explanation pass, and one delayed revision pass. Keep an error log organized by contract, invariant, numeric safety, data-structure choice, complexity, and test coverage. A roadmap edition establishes ordering and scope; return as its chapter set expands rather than treating it as finished instruction.

# Contents

This contents page covers only the current focused PDF. Section-level navigation is available through PDF bookmarks.

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# Series Roadmap

Choose Java, DSA, Frameworks, or System Design first, then follow the books inside that segment in order. The same series codes and positions appear on the website, PDF covers, and canonical download folders.

| SEGMENT    | BOOK           | FOCUSED LEARNING PATH                                     |
|------------|----------------|-----------------------------------------------------------|
| Java       | <b>JAVA 01</b> | Java Foundations for Problem Solving                      |
| Java       | <b>JAVA 02</b> | Git and GitHub for Java Engineers                         |
| Java       | <b>JAVA 03</b> | Maven and Gradle for Java Engineers                       |
| Java       | <b>JAVA 04</b> | Language, OOP, and Modern Java                            |
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| Java       | <b>JAVA 07</b> | Java Concurrency and the Memory Model                     |
| Java       | <b>JAVA 08</b> | Performance, Diagnostics, and GC Incidents                |
| Java       | <b>JAVA 09</b> | Java Question Bank, Study Plan, and Reference             |
| DSA        | <b>DSA 01</b>  | Time and Space Complexity for Java Interviews             |
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| DSA        | <b>DSA 05</b>  | Loop Mastery, Patterns, and Index Calculations            |
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| Frameworks | <b>FW 03</b>   | Spring Framework for Java Engineers                       |
| Frameworks | <b>FW 04</b>   | Spring Boot for Java Backend Engineers                    |
| Frameworks | <b>FW 05</b>   | Spring Boot and REST APIs                                 |

| SEGMENT       | BOOK  | FOCUSED LEARNING PATH                  |
|---------------|-------|----------------------------------------|
| Frameworks    | FW 06 | Spring Data for Java Backend Engineers |
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All 40 publication editions are available now. Choose Java, DSA, Frameworks, or System Design first, then follow the books inside that shelf in order. Each deep-dive book remains independently printable and available on the web.

# About the Author

## Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer and the author and editor of the Java SDE-2 Interview Preparation Series. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

## Editorial approach

Vinay sets the learning sequence and publication standard and performs the final review for Java accuracy, evidence, attribution, and release readiness. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

## Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

## Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

**Connect:** [LinkedIn](#) | [GitHub](#)

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Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: Series Index - 4 Segments / 41 Books. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

Source provenance: curated from the independent Java SDE-2 master guide plus focused original material. Original master chapter numbers are labeled explicitly when retained in cross-references.